

TEGYULTYA, Russian Federation

WMO: 249670

Lat: 60.467N

Lon: 130.000E

Elev: 562

StdP: 14.40

Time Zone: 9.00 (E09)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-55.7	-52.7	-59.0	0.2	-55.3	-56.0	0.2	-52.5	9.7	-16.9	8.3	-16.5	0.1	270	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.1	86.6	67.1	82.8	65.1	79.1	63.6	70.0	81.6	67.7	78.2	65.4	75.6	4.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	99.0	74.9	63.8	90.6	73.1	61.4	83.0	70.5	34.4	82.2	32.6	78.4	30.6	76.1	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	9.9	8.4	DB	-59.3	91.8	3.1	4.5	-61.5	95.0	-63.3	97.6	-65.1	100.1	-67.3	103.4	(4)
(5)			WB	-59.2	76.2	2.9	5.3	-61.3	80.0	-63.0	83.1	-64.7	86.0	-66.8	89.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	-32.6	-26.6	-2.9	23.0	45.7	58.8	64.1	58.1	41.4	20.2	-14.8	-31.9		
	DBStd	37.15	11.45	11.33	13.25	11.85	7.06	6.27	6.31	6.81	7.58	11.35	14.61	11.57		
	HDD50	13028	2560	2146	1641	811	169	7	1	12	273	925	1944	2539		
	HDD65	17576	3025	2566	2106	1261	597	201	92	232	708	1390	2394	3004		
	CDD50	1025	0	0	0	0	37	270	439	264	15	0	0	0		
	CDD65	98	0	0	0	0	0	14	65	19	0	0	0	0		
	CDH74	2089	0	0	0	0	28	526	1119	411	5	0	0	0		
	CDH80	659	0	0	0	0	3	130	404	122	0	0	0	0		
	WSAvg	2.7	1.2	1.4	2.5	3.4	4.2	3.4	3.3	3.0	3.4	3.2	2.2	1.6		
	PrecAvg	11.5	0.5	0.4	0.3	0.4	1.1	1.3	1.9	2.0	1.6	0.9	0.7	0.5		
	PrecMax	17.2	1.2	0.8	0.7	1.3	2.8	3.1	3.8	5.0	3.7	2.6	2.6	0.9		
(15) Precipitation	PrecMin	7.6	0.1	0.1	0.1	0.0	0.2	0.3	0.4	0.3	0.5	0.3	0.2	0.2		
	PrecStd	2.6	0.3	0.2	0.2	0.3	0.6	0.7	1.1	1.2	0.8	0.5	0.5	0.2		
	DB	5.5	12.0	43.5	59.3	79.0	87.2	95.6	89.0	72.4	46.4	33.4	7.3			
	MCWB	4.8	11.1	35.8	43.3	58.0	64.7	71.4	68.2	57.8	41.7	31.0	6.8			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	-2.6	4.5	35.1	52.9	72.6	83.8	88.8	83.1	65.3	41.7	24.0	-2.9			
	2% MCWB	-3.1	3.6	28.8	40.6	52.7	63.2	69.2	66.1	54.0	37.0	22.3	-3.5			
	5% DB	-9.3	-2.0	30.3	48.4	68.3	80.2	84.4	78.6	60.5	38.2	14.7	-7.5			
	5% MCWB	-9.8	-2.8	25.3	37.5	51.1	62.1	67.3	64.0	51.0	34.7	13.7	-7.9			
	10% DB	-15.3	-7.4	23.7	44.0	63.3	76.5	80.7	74.1	56.1	35.4	6.6	-13.3			
	10% MCWB	-15.7	-8.0	19.9	35.2	48.6	60.4	65.6	61.6	48.4	33.0	5.9	-13.2			
	0.4% WB	4.5	10.9	35.8	45.2	59.8	68.9	75.8	72.0	58.7	42.0	31.3	6.9			
	0.4% MCDB	5.1	11.6	42.4	56.7	77.5	80.9	89.0	82.8	70.0	45.9	32.6	7.1			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	-3.0	3.6	29.5	41.0	54.6	65.9	71.3	68.6	56.0	38.2	22.3	-3.4			
	2% MCDB	-2.5	4.3	34.7	51.8	67.6	78.5	84.4	79.0	64.0	40.4	23.8	-3.0			
	5% WB	-9.8	-2.7	25.3	38.3	52.0	63.7	69.2	65.8	52.6	35.4	13.7	-7.9			
	5% MCDB	-9.4	-2.1	30.2	47.6	64.4	76.3	80.4	75.2	58.6	37.0	14.7	-7.5			
	10% WB	-15.7	-7.9	20.2	35.5	49.8	61.6	67.2	62.9	49.3	32.9	5.9	-13.5			
	10% MCDB	-15.4	-7.3	23.9	43.4	60.7	74.1	77.4	72.6	54.7	35.6	6.6	-13.2			
	MDBR	16.7	26.8	36.7	32.1	25.2	27.3	26.1	25.9	23.2	17.2	19.5	14.2			
	5% DB	22.6	31.6	40.2	32.9	36.7	36.5	36.2	35.4	31.3	17.9	25.7	20.6			
(36) Mean Daily Temperature Range	MCWBR	22.2	30.6	34.1	21.1	19.1	18.2	18.8	19.1	20.0	13.5	24.4	20.1			
	MCDBR	22.4	31.6	38.5	31.2	30.0	31.4	29.6	30.1	26.9	14.7	24.9	20.6			
	MCWBR	22.1	30.7	32.8	20.7	17.0	16.7	16.8	17.0	17.9	11.9	23.8	20.1			
	MCDBR	22.1	30.7	32.8	20.7	17.0	16.7	16.8	17.0	17.9	11.9	23.8	20.1			
(40) Clear-Sky Solar Irradiance	taub	0.232	0.271	0.309	0.376	0.453	0.427	0.509	0.460	0.359	0.298	0.239	0.205			
	taud	1.986	2.047	2.041	2.002	2.030	2.232	1.981	2.090	2.347	2.287	2.016	1.912			
	Ebn at Noon	160	223	253	255	243	255	226	227	239	216	153	117			
	Edh at Noon	14	25	36	45	48	40	50	42	27	20	14	10			
(44) All-Sky Solar Radiation	RadAvg	156	490	1035	1620	1651	1857	1696	1331	842	421	208	83			
	RadStd	9	19	37	59	152	130	121	100	58	46	14	3			

Historical Trends

DBAvg	Heating		Cooling			Degree-Days			
	99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	+1.26	+2.48	+2.32	N/A	N/A	N/A	-428	-432	N/A

Nomenclature: See separate page